



Master Presenter's Manual

for

Data Management with SQL Server

Curriculum Code: OV-1507

***Effective from: March, 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Describe the architecture and editions of SQL Server.
- Work with SQL Server Management Studio.
- Describe the basic architecture, versions, and editions of SQL Server.
- Describe the role of SQL Azure compared to on-premises SQL Server.
- Describe how to connect to SQL Azure with SSMS.
- Categorize SQL statements into their dialects.
- Identify the elements of T-SQL, including predicates, operators, expressions, and comments.
- Describe the elements of a SELECT statement.
- Write SELECT DISTINCT clauses.
- Describe the common join types in T-SQL queries.
- Write queries that use joins.
- Describe how to sort and filter data.
- Describe the difference between implicit and explicit data type conversion.
- Describe the character data types supplied by SQL Server.
- Describe the impact of collation on character data.
- Concatenate strings.
- Extract and manipulate character data using built-in functions.
- Write queries comparing dates and times.
- Write queries that use the HAVING clause to filter groups.
- Use Table Expressions.
- Use Set Operators.
- Write queries that use window aggregate, window ranking, and window offset functions.
- Describe the architecture of SQL Server.
- Describe how to configure the number of tempdb data files.
- Explain Multiple TempDB data files.
- Outline the Execution Plans feature in SQL Server.
- Explain how to automatically keep the history of the data in the table.
- Explain how to protect data at rest and in motion.
- Describe Security enhancements in SQL Server.
- Explain how to work with JSON data.
- Explain how to work with PolyBase.
- Define and describe Query Store.
- Explain how to dynamically stretch warm and cold transactional data from SQL Server to Azure.
- Describe how to tune workload performance with Query Store.
- Learn the new Intelligent Query Processing (IQP) features in SQL Server 2022.
- Learn the features of Machine Learning Services.

2. Information on Module Allocation

Module	Online (No. of Hrs)	Skill Code
SQL	32	1291

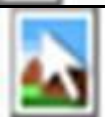
Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 SQL-TL1	Session 2 SQL-TL2	Session 3 SQL-TL3	Session 4 SQL-TL4
2	Session 5 SQL-TL5	Session 6 SQL-TL6	Session 7 SQL-TL7	Session 8 SQL-TL8
3	Session 9 SQL-TL9	Session 10 SQL-TL10	Session 11 SQL-TL11	Session 12 SQL-TL12
4	Session 13 SQL-TL13	Session 14 SQL-TL14	Session 15 SQL-TL15	Session 16 SQL-TL16

SQL: Data Management with SQL Server

TL: Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	SQL – TL1	All the topics as listed below from Session 1 and Session 2 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 1 – RDBMS CONCEPTS</u> ➤ Explain the concept of data and database ➤ Describe the approaches to data management ➤ Define a Database Management System (DBMS) and list its benefits ➤ Explain different database models ➤ Define and explain RDBMS ➤ Describe entities and tables and list the characteristics of tables ➤ List the differences between a DBMS and an RDBMS <u>Session 2 – Entity-Relationship (E-R) Model and Normalization</u> ➤ Define and describe data modeling ➤ Identify and describe the components of the E-R model ➤ Identify relationships that can be formed between entities ➤ Explain E-R diagrams and their use ➤ Describe an E-R diagram, the symbols used for drawing, and show various relationships ➤ Describe various Normal Forms ➤ Outline uses of different Relational Operators	<u>SQL Server - The Definitive Guide</u> SG - Session 1 & 2 XP - Session 1 & 2 TG - Session 1 & 2
2	SQL – TL2	All the topics as listed below from Session 3 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 3 – Introduction to SQL Server 2022</u> ➤ Describe an overview of SQL Server 2022	<u>SQL Server - The Definitive Guide</u> SG - Session 3 XP - Session 3 TG - Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain basic architecture and version history of SQL Server 2022 ➤ Identify new features added to SQL Server 2022 ➤ Outline the process of connecting to SQL Server instances ➤ Define databases and list the key features of AdventureWorks 2022 sample database ➤ Explain the components of SQL Server Management Studio (SSMS) ➤ Elaborate on script file creation and organization ➤ Explain the process to execute Transact-SQL queries	
3	SQL – TL3	All the topics as listed below from Session 4 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 4 – Transact-SQL</u> ➤ Explain Transact-SQL ➤ List different categories of Transact-SQL statements ➤ Explain various data types supported by Transact-SQL ➤ Explain Transact-SQL language elements ➤ Explain sets and predicate logic ➤ Describe logical order of operators in the SELECT statement	<u>SQL Server - The Definitive Guide</u> SG - Session 4 XP - Session 4 TG - Session 4
4	SQL – TL4	The Try It Yourself questions of Session 1 to Session 4 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session.	<u>SQL Server - The Definitive Guide</u> Session 1-4
5	SQL – TL5	All the topics as listed below from Session 5 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 5 – Creating and Managing Databases</u> ➤ Explain about modification of system data ➤ Describe adding of filegroups and	<u>SQL Server - The Definitive Guide</u> SG - Session 5 XP - Session 5 TG - Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
		transaction logs ➤ Outline the process to create a database ➤ Describe how to drop a database ➤ Explain database snapshots	
6	SQL – TL6	All the topics as listed below from Session 6 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 6 – Creating Tables</u> ➤ List SQL Server 2022 data types ➤ Describe the procedure to create, modify, and drop tables in an SQL Server database ➤ Describe the procedure to add, modify, and drop columns in a table	<u>SQL Server - The Definitive Guide</u> SG - Session 6 XP - Session 6 TG - Session 6
7	SQL – TL7	All the topics as listed below from Session 7 and Session 8 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 7 – Microsoft Azure SQL</u> ➤ Explain Azure SQL ➤ List the features and benefits of Azure SQL ➤ State the differences between Azure SQL and on-premises SQL Server ➤ Explain steps to connect Azure SQL with SSMS <u>Session 8 – Accessing Data</u> ➤ Describe SELECT statement, its syntax, and use ➤ Explain various clauses used with SELECT ➤ State the use of ORDER BY clause ➤ Describe working with typed and untyped XML ➤ Explain the procedure to create, use, and view XML schemas	<u>SQL Server - The Definitive Guide</u> SG - Session 7 & 8 XP - Session 7 & 8 TG - Session 7 & 8

Session No.	Session Title	Session Details	Deliverables' Mapping
8	SQL – TL8	The Try It Yourself questions of Session 5 to Session 8 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session.	<u>SQL Server - The Definitive Guide</u> Session 5-8
9	Q	All the topics as listed below from Session 9 and Session 10 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 9 – Advanced Queries and Joins</u> ➤ Explain grouping and aggregating data ➤ Describe subqueries ➤ Describe table expressions ➤ Explain Joins ➤ Describe various types of Joins ➤ Explain the use of various set operators to combine data ➤ Describe pivoting and grouping set operations <u>Session 10 – Using Views, Stored Procedures, and Querying Metadata</u> ➤ Define views ➤ Describe the technique to create, alter, and drop views ➤ Define stored procedures and its types ➤ Describe the procedure to create, alter, and execute stored procedures ➤ Describe nested stored procedures ➤ Describe querying SQL Server metadata ➤ System Catalog views and functions	<u>SQL Server - The Definitive Guide</u> SG - Session 9 & 10 XP - Session 9 & 10 TG - Session 9 & 10
10	SQL – TL10	All the topics as listed below from Session 11 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 11 – Indexes</u> ➤ Define and explain Indexes ➤ Explain Storage Structure ➤ Explain types of Indexes	<u>SQL Server - The Definitive Guide</u> SG - Session 11 XP - Session 11 TG - Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Elaborate on Index Management	
11	SQL – TL11	All the topics as listed below from Session 12 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 12 – Triggers</u> ➤ Explain triggers ➤ Explain the procedure to create and alter DML triggers ➤ Describe nested triggers ➤ Describe update functions ➤ Explain the handling of multiple rows in a session ➤ Explain the performance implication of triggers implication	<u>SQL Server - The Definitive Guide</u> SG - Session 12 XP - Session 12 TG - Session 12
12	SQL – TL12	The Try It Yourself questions of Session 9 to Session 12 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session.	<u>SQL Server - The Definitive Guide</u> Session 9-12
13	SQL – TL13	All the topics as listed below from Session 13 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 13 – Programming Transact-SQL</u> ➤ Explain Transact-SQL programming ➤ Identify program flow statements ➤ Describe various Transact-SQL functions ➤ Explain the procedure to create and alter User-Defined Functions (UDFs) ➤ Elaborate on creation of windows with OVER ➤ Define window functions	<u>SQL Server - The Definitive Guide</u> SG - Session 13 XP - Session 13 TG - Session 13
14	SQL – TL14	All the topics as listed below from Session 14 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 14 – Transactions and Error Handling</u>	<u>SQL Server - The Definitive Guide</u> SG - Session 14 XP - Session 14

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Define and describe transactions ➤ Explain the procedure to implement transactions ➤ Explain the steps to mark a transaction ➤ Distinguish between implicit and explicit transactions ➤ Explain isolation levels ➤ Explain the scope and different types of locks ➤ Explain transaction management ➤ Explain error handling and its implementation ➤ Elaborate on the TRY-CATCH block ➤ Explain the procedure to display error information ➤ Describe the @@ERROR and RAISERROR statements ➤ Explain the use of ERROR_STATE, ERROR_SEVERITY, and ERROR_PROCEDURE ➤ Explain the use of ERROR_NUMBER, ERROR_MESSAGE, and ERROR_LINE ➤ Elaborate on the THROW statement	TG - Session 14
15	SQL – TL15	All the topics as listed below from Session 15 and Session 16 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session. <u>Session 15 – PolyBase and Query Store</u> ➤ Describe PolyBase ➤ Explain features and advantages of PolyBase ➤ Define and describe Query Store ➤ Describe how to tune workload performance with Query Store <u>Session 16 – Intelligence and Machine Learning in SQL Server 2022</u> ➤ Describe Intelligent Query Processing (IQP) ➤ Describe Parameter Sensitive Plan (PSP)	<u>SQL Server - The Definitive Guide</u> SG - Session 15 & 16 XP - Session 15 & 16 TG - Session 15 & 16

Session No.	Session Title	Session Details	Deliverables' Mapping
		optimization ➤ Explain the features of Machine Learning Services ➤ List the steps for installation of SQL Server 2022 Machine Learning Services to run Python scripts ➤ Explain the steps to predict future data with linear regression and SQL Machine Learning	
16	SQL – TL16	The Try It Yourself questions of Session 13 to Session 16 of <i>SQL Server-The Definitive Guide</i> book should be covered in this session.	<u>SQL Server - The Definitive Guide</u> Session 13-16

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Student:

Module	Deliverable
Data Management with SQL Server	Kit Code: OV-1507-KIT01-01-A Kit Name: Kit of 1 title for OV-1507 Data Management SQL Ser

Deliverables for Faculty:

Module	Deliverable
Data Management with SQL Server	Kit Code: OV-1507-KIT01-01-A Kit Name: Kit of 1 title for OV-1507 Data Management SQL Ser
Course	Classroom Presentations Trainer Guide Master PM

6. Library References

➤ Murach's SQL Server 2022 for Developers by Bryan Syverson
➤ T-SQL Fundamentals, 4th Edition by Itzik Ben-Gan
➤ SQL Server 2022 Revealed: A Hybrid Data Platform Powered by Security, Performance, and Availability by Bob Ward

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for **PASS**
Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for **CREDIT** and
Overall Weighted Marks $\geq 75\%$ qualifies for **DISTINCTION**

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination (E1)

Marks: 50
Duration: 60 minutes
Type: Objective

Sr. No.	Name of the Subject	Distribution of Marks
1.	Data Management with SQL Server	50

Final Examination (R1)

Marks: 50
Duration: 120 minutes
Type: Practical

Sr. No.	Name of the Subject	Distribution of Marks
1.	Data Management with SQL Server	50

Evaluation Details

Practical – 50%
OES – 50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the module of this course have been described as follows:

8.1 Software Requirements

Microsoft Windows 10 or higher
Microsoft SQL Server 2022

8.2 Recommended Hardware

Intel Core i3/i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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